

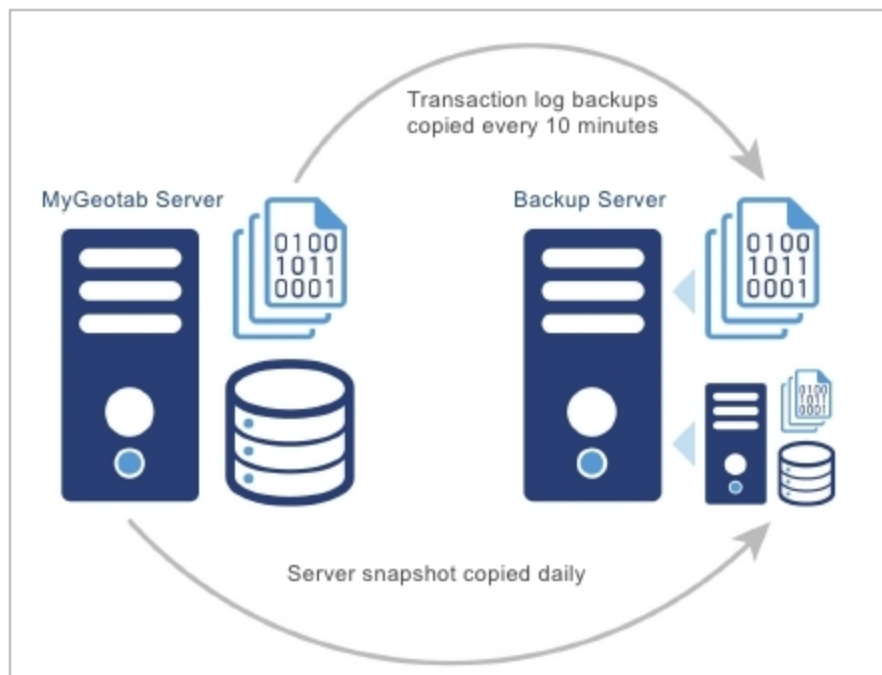


Figure 2: How a backup solution creates constant backups
(image source: [googleimages.com](https://www.google.com/search?q=googleimages.com))

various clients in such a way that the total backup time is about the same for every backup run.

9. It frees system admins from having to guess the rate of data changes in their environments.

BackupPC

This is a free disk-to-disk backup software suite with a Web-based front-end. This cross-platform server can run on any Solaris, Linux or UNIX based server. In 2007, it was ranked among the three most well-known open source backup software. Even though it's so amazing, not many have heard of it. This network backup system can archive a large number of files to a local or networked disk storage solution. Data de-duplication actually reduces the disk space that is required to store the backups in the disk pool. It is possible to use BackupPC as a disk-to-disk-to-tape solution, if its archive function is used to back up the disk pool to tape.

Features:

1. BackupPC uses compression and pooling to make different archived files as small as possible, hence reducing the storage hardware capacity costs and requirements.
2. No dedicated client is required as the server itself acts as a client for several protocols that are being handled by other services native to the client OS.
3. It is not a block-level backup system, but performs file based restore and backup.
4. It incorporates a SMB (server message block) client that can be used to back up the network shares of computers running Windows.
5. It is considered very useful for Web servers running SSH (secure shell) with GNU tar and rsync available, as it allows the BackupPC server to be stored in a subnet that is separate from the Web server's DMZ.
6. It supports Windows, Solaris, Linux or UNIX based servers.

Bacula

This is basically a set of computer programs used for recovery, backup and verification of computer data across a network of different kinds of computers. It's relatively easy and efficient to use. It offers a couple of advanced storage management features that make it simple to find and recover the lost or damaged files. It's made up of five major components or services.

1. The Bacula Director service actually supervises different backups, and verifies, restores and archives various operations.
2. The Bacula Console service lets the administrator or user communicate with the Bacula Director. Currently, the Bacula Console is available in three versions: GNOME-based interface, text-based console interface and a wxWidgets graphical interface.
3. The Bacula File service (also referred to as the client program) is generally installed on the machine to be backed up.
4. The Bacula Storage service consists of various software programs that perform storage and recovery of the file attributes as well as data to the physical backup media.
5. The Catalogue service comprises different software programs responsible for maintaining the file indexes and volume databases for all the backed up files.

Features:

1. Supports Windows, Linux and UNIX based OSs.
2. Restores and backs up clients of any type, ensuring that all the attributes of the files are properly restored and saved.
3. Supports multi-volume backups.
4. Provides a comprehensive SQL standard database for all the files that are backed up. This allows online viewing of the saved files on any particular volume.
5. Offers enterprise-grade recovery, backup and data verification capabilities.

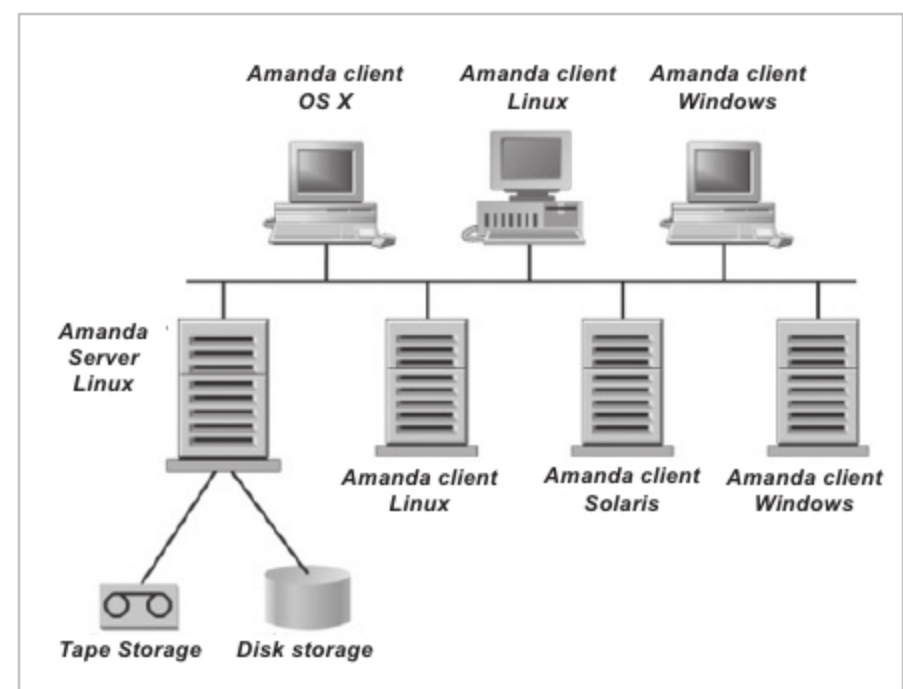


Figure 3: Architecture for the AMANDA backup solution
(image source: [googleimages.com](https://www.google.com/search?q=googleimages.com))

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